S. V. National Institute of Technology, Surat

Applied Mathematics and Humanities Department

B.Tech–I Sem–1 Branch–All Subject–Mathematics-I (MA 101 S1)

Tutorial - 7 : Curve Tracing:Cartesian, polar and parametric form of standard curves.

(1) Draw the circle $x^2 + y^2 - 4x + 4y + 4 = 0$.

(2) Sketch the following curves:

i. $y^2 + 4x - 4y + 8 = 0$.

ii. $x^2 + 4x - 4y + 16 = 0$.

(3) Trace the curve: $y^2(2a - x) = x^3$.

(4) Trace the curve: $y^2x = a^2(a - x)$.

(5) Trace the curve: $x^{1/2} + y^{1/2} = a^{1/2}$.

(6) Trace the curve: $y = 2x + \frac{x^2}{2} - \frac{x^3}{3}$

(7) Trace the curve: $r = a(1 + cos\theta)$

(8) Trace the curve: $r = ae^{\theta \cot \alpha}$; $\alpha > 0$, (Equiangular spiral)

(9) Trace the curve: $r(1 + cos\theta) = 2a$ (Parabola opening on left)

(10) Trace the curve: $x = acost + \frac{1}{2}\log tan^2\left(\frac{t}{2}\right)$, $y = asint$

(11) Sketch the following curves:

i. $x = asin2\theta(1 + cos2\theta)$, $y = acos2\theta(1 - cos2\theta)$

ii. $r = \frac{sin^2\theta}{cos\,\theta}$

(12) Trace the curve: $x = a(t + sint)$, $y = a(1 + cost)$

(13) Trace the curve: $x = acos^3t, y = asin^3t$ $(Cartesian\,form: x^{2/3} + y^{2/3} = a^{2/3})$

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